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MDA and PASS system for ArRass2 Solar PV IPP in Saudi Arabia

CLIENT NAME:

LARSEN & TOUBRO LIMITED LNTECC

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TECHNICAL PROPOSAL



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Plant Performance Model and Plant Accounting Settlement System Software for the ArRass 2 Solar PV IPP



1.0 Proposal Summary

1.1 General

MEMF ITS Ideal Technical Solutions (MEMF ITS) is bidding as a supplier for the combined metering data acquisition and settlement system scope to the EPC Larson & Toubro of India, for the Ar Rass 2 Solar PV Independent Power Plant (Ar Rass 2 PV IPP). The Power Purchase Agreements (PPA) for this IPP foresee complex regulations for the settlement of the transactions, including performance declarations to be evaluated by means of Plant Performance Models (PPM).

ENEXSA GmbH (ENEXSA, Seller) proposes to provide the services to define, configure and commission the Plant Performance Model and Settlement System Software in form of ENEXSA's PASS Plant Accounting and Settlement System Software and related Remote Services as described in this Proposal.

1.2 Plant Scope Basis

The scope of the Ar Rass 2 project comprises of a solar PV power plant with a total net capacity of 2000 MWe which will be located in the Riyadh province, Saudi Arabia. The electricity generated by this plant will be purchased by the offtaker Saudi Power Procurement Company (SPPC) under a 25-year Power Purchase Agreement (PPA).

1.0 Proposal Summary

1.3 Deliverables

Deliverables of the works offered in this Proposal will be the fully engineered, configured, and commissioned Plant Performance Model and Plant Accounting and Settlement System Software as further described in Section 3 of this Proposal. The scope of supply also includes remote training services for plant personnel and SPPC staff in the use and administration of the various software systems involved.

Supply of one copy of the PPM and PASS software (for synchronized mode, “Offline PSS”) for installation at SPPC offices on server hardware supplied by Buyer is also included in the scope of supply.

2.0 EXPERIENCE LIST

Plant	Year	Country	Scope	Owner/Developer
Fujairah I IWPP	2008	UAE	Reference HB, FDM and PASS	Emirates SembCorp
Fujairah II IWPP	2008	UAE	Reference Heat Balances	SembCorp/Soijtz/Kyushu
Marafiq Jubail IWPP	2008	KSA	PASS	Suez/GIC/ACWA
Salalah IWPP	2010	Oman	Reference Heat Balances, FDM	SembCorp
Shuweihat S2 Power Water Plant	2011	UAE	FDM & PASS	IPR GDF SUEZ
Al Dur Power and Water Plant	2011	Bahrain	FDM & PASS	IPR GDF SUEZ
Barka 3 / Sohar 2 IPP	2010	Oman	Reference Heat Balances	Marubeni
Shuweihat S3 IPP	2010	UAE	Reference Heat Balances	Marubeni
Sur IPP	2013	Oman	Reference HB, FDM and PASS	Marubeni/QWEC
Hassyan IWPP	2012	UAE	Reference Heat Balances	Marubeni/TAQA
Shams 1 Solar IPP	2012	UAE	PASS	TOTAL/Abengoa Solar
Shuweihat S3 IPP	2013	UAE	FDM & PASS	Sumitomo
Mirfa IWPP	2013	UAE	Reference Heat Balances	Marubeni
Saudi Aramco Cogen Projects	2013	KSA	Reference Heat Balances	Marubeni
Al Ghubrah Water Plant	2015	Oman	PASS	Muscat City Desal.Co.
Az-Zour North Phase 1 IWPP	2015	Kuwait	FDM & PASS	Engie, Sumitomo, Al Sagar
Noor 1 Solar IPP	2015	Morocco	PASS	NOMAC (ACWA Power)
Mirfa IWPP	2016	UAE	FDM & PASS	GDF SUEZ
KAHRAMAA Facility D	2015	Qatar	Reference Heat Balances	Mitsubishi Corporation
Al Dhafra/Al Dhafra IPP	2015	Oman	Reference Heat Balances	Marubeni
Az-Zour North Phase 2 IWPP	2016	Kuwait	Reference Heat Balances	Marubeni, Alghanim
Musandam IPP	2016	Oman	PASS	Wärtsilä
RAWEC ISPP	2016	KSA	Fuel Demand Model	RAWEC
Qurayyat IWP	2017	Oman	PASS	Hyflux
Salalah 2 IPP	2017	Oman	FDM & PASS	Mitsui, ACWA Power
Myingyan IPP	2017	Myanmar	FDM & PASS	Sembcorp
Barka IWP	2018	Oman	PASS	Engie, SUEZ International
Noor 2 Solar IPP	2018	Morocco	PASS	ACWA Power
Noor 3 Solar IPP	2018	Morocco	PASS	ACWA Power
Fadhili CHP	2018	KSA	FDM & PASS	Engie
Sweihan PV IPP	2018	UAE	PASS	Marubeni
Sirajganj S4 IPP	2019	Bangladesh	FDM & PASS	Sembcorp
KAHRAMAA Facility D	2019	Qatar	FDM, ADM & PASS	Mitsubishi Corporation
Sakaka PV IPP	2019	KSA	PPM & PASS	ACWA Power



2.0 EXPERIENCE LIST - Continue

Plant	Year	Country	Scope	Owner/Developer
Salalah IWP	2020	Oman	PASS	ACWA Power
Shuqaiq 3 IWP	2021	KSA	PASS	Marubeni
Mesaieed IWPP	2021	Qatar	PASS (Replacement)	Marubeni
Rabigh 3 IWP	2021	KSA	PASS	ACWA Power
Fujairah F3 IPP	2021	UAE	Reference Heat Balances	Marubeni
Saudi Aramco Tanajib ISPP	2021	KSA	Reference Heat Balances	Marubeni
Taweelah A1 IWPP	2022	UAE	PASS (Replacement)	Total
Facility E IWPP	2022	Qatar	Reference Heat Balances	Marubeni
Hamriyah IPP	2022	UAE	FDM & PASS	Sumitomo
Al Dhafrah PV2 IPP	2022	UAE	PASS	EDF Renouvables
Umm al Quwain IWP	2022	UAE	PASS	ACWA Power
KAHRAMAA Al Kharsaa Solar PV IPP	2023	Qatar	SPM & PASS	Marubeni/Total
Jubail 3A IWP	2023	KSA	PASS	ACWA Power



■ 3.0 TECHNICAL PROPOSAL

3.1 PPM & PASS Software

3.1.1 Plant Performance Model (PPM)

The PPM will be designed to meet the requirements of the Power Purchase Agreement (PPA) of the IPP.

The Plant Performance Model shall provide a mathematical representation of the Plant, including its technical configuration and shall be designed to provide the following outputs, including but not limited to:

- o Actual performance ratio;
- o Projected performance ratio;
- o Actual GPOA versus expected GPOA
- o Comparison of Projected Net Electrical Energy versus Net Electrical Energy;
- o Coefficients A, B and C of the Deemed Electrical Energy Payment; and
- o Forecast values for Projected Net Electrical Energy.

The following inputs to the Plant Performance Model shall be fed online:

- o Global horizontal irradiation;
- o Direct normal irradiation;
- o Diffuse irradiation;
- o Global solar irradiation on plane of array;
- o Ambient air temperature;
- o Module temperature;
- o Power factor;
- o Net Electrical Energy; and
- o Other data, as applicable.

Figure 1: Excerpt from a General Technical Specification describing the Plant Performance Model

Clarification: The Projected Performance Ratio is not calculated by the Plant Performance Model. It is already specified in the PPA itself (separately for each Contract Year). The PPM can compare the Actual Performance Ratio (based on measured data) with this contractual guarantee.

Clarification: The values for the above list of online inputs for the PPM are recorded for reference only.

The Actual Performance Ratio will be calculated based on:

- Net Electrical Energy
- Module Temperature, and
- Global Solar Irradiation on Plane of Array (GPOA)

While the PASS will include a Regression Tool to determine the coefficients A, B, and C for the calculation for Deemed Electrical Energy Payment as per Schedule 17 of the PPA, ENEXSA explicitly excludes forecasting calculations from its scope of supply. The PPM as a forecasting tool is not related to the tariff equations of the PPA and such calculations can be performed by an external tool utilizing weather forecast data available in the market.

Forecast data can be imported into the PASS in a mutually agreed format to be specified in the Functional Design Specification for the PASS (see Section 3.2.2).

The PASS is capable of processing all on-line input data with a granularity of one (1) minute as required by the SPPC specification, but for practicality of the day-to-day work with the PASS, ENEXSA suggests using 6-minute time intervals. The granularity of the input data shall be agreed with the project company in the Functional Design Specification for the PASS (see Section 3.2.2).

In a third-party audit, a Technical Consultant shall verify that the PPM performs accurately all functions required by the PPA. ENEXSA will support the offline audit of the PASS and PPM with services as specified in Section 3.2.4.

3.1.2 Plant Accounting and Settlement System (PASS)

The PASS settlement system will be designed for calculation of tariffs and the monthly generation of invoices as defined in Schedule 9 the Power Purchase Agreement (PPA) of the IPP.

The plant settlement system shall consist of several modules that will perform separate tasks for data collection, archiving and processing, and invoice generation. All modules will reside on hardware to be supplied by Buyer. Figure 2 below shows the interface schematic of such system setup. The data exchange between the various SQL data bases on-site and at the SPPC CSS shall be via ODBC. The respective table structures for writing to the CSS Data Base must be provided by SPPC. The synchronization of all automated and manual system inputs between the settlement system at the plant site and the system copy at the CSS will be accomplished via Advanced Message Queue Protocol (AMQP).

Validation of all input data (including processing of main and check meters) takes place on an hourly basis, and the result of the automatic validation can be viewed and – if required – modified in the respective dialogs of the system.

The following software modules are part of the PASS system:

- Dareios SV Data Base Module
- Dareios CN Calculation Node
- Dareios RGM Report Generation Module
- Dareios IGM Invoice Generation Module
- Dareios RT Regression Tool
- Dareios SmartClient

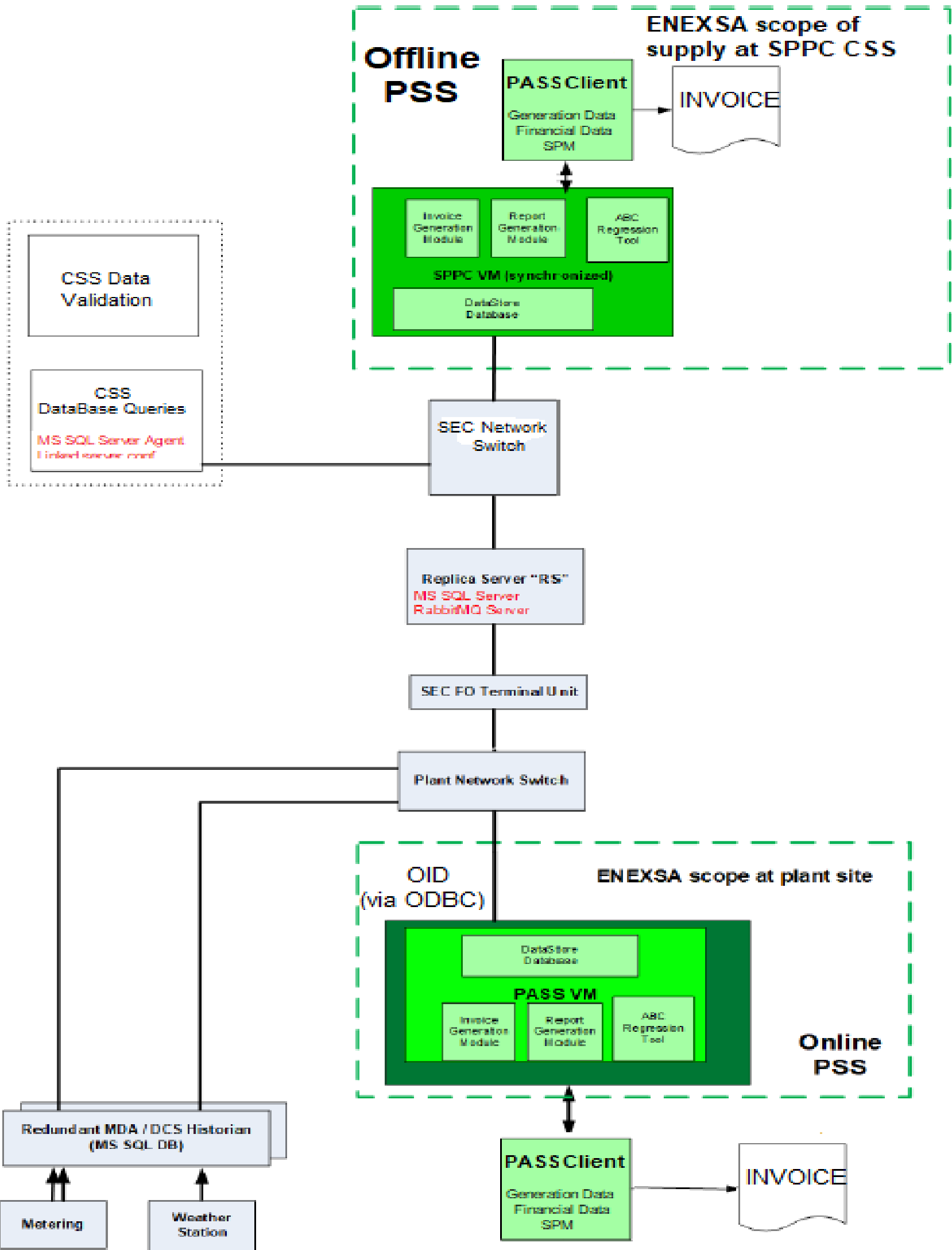


Figure 2 Interaction of PASS software with the settlement infrastructure for synchronization with SPPC



3.1.2.1 Dareios SV Data Base Module

Microsoft SQL Server will be used as the database for the on-line PPM and Settlement System. Security and user management are standard features of the Microsoft SQL Server. The SQL server will manage the Dareios DB database (internal database) for archiving of the PPM and settlement system data.

3.1.2.2 Dareios CN Calculation Node

The Dareios CN Calculation Node software manages the data I/O, the data pre-processing and the execution of the Wind Plant Performance Model.

3.1.2.3 Dareios IGM Invoice Generation Module

The Invoice Generation Module will be designed to meet the requirements of the PPA and all related contractual documents. With additional user inputs for financial data, it will process the plant data generated by the metering system and the PPM, and it will calculate all charge rates as per the definitions of the PPA.

3.1.2.4 Dareios RGM Report Generation Module

Through various standard reports, the Plant Accounting System enables the users to quickly evaluate plant performance relative to the PPA guarantees at any given point of time in the past and over varying periods.

Reports cover all aspects of the invoicing process, such as:

- Generation data,
- Contractual data,
- Monthly/period data, and
- Availability information.

3.1.2.5 Dareios RT Regression Tool

The Deemed Electrical Energy of the PPA is calculated by a second-order polynomial of the global horizontal irradiation GHI with three parameters A, B, and C that have to be derived for every month of the year by regression from actual production data of the previous year.

The Regression Tool provides inputs for the user to define the following limit parameters for the regression analysis:

- **$E_{k,act}$** ... minimum electrical energy below which data shall not be considered for the regression
- **$GPOA_k$** ... minimum value of global solar irradiation on the plane of array below which data shall not be considered for the regression
- **AVL_k** ... minimum value of plant availability below which data shall not be considered for the regression
- **AVL Flag** ... user input to turn on/off the correction to 100% availability in the regression at times of reduced availability

Upon execution of the regression, the user can copy the resulting values for the coefficients A, B, and C to the respective input fields for the respective month of the following year. This process is intentionally designed as a manual interaction, since the parameters may also be altered by the user for other reasons.

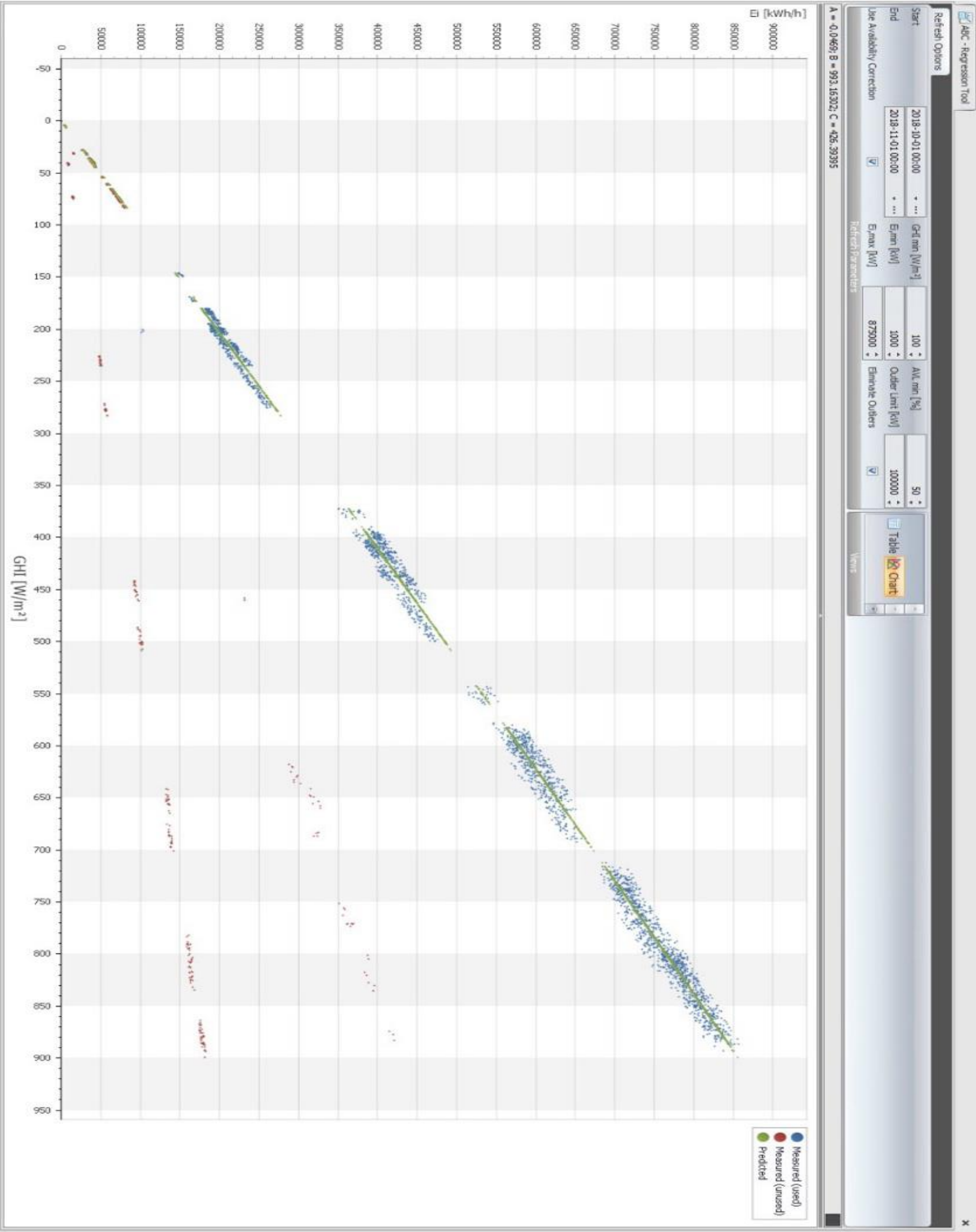


Figure 3: Scatter plot of the Regression Tool for data analysis

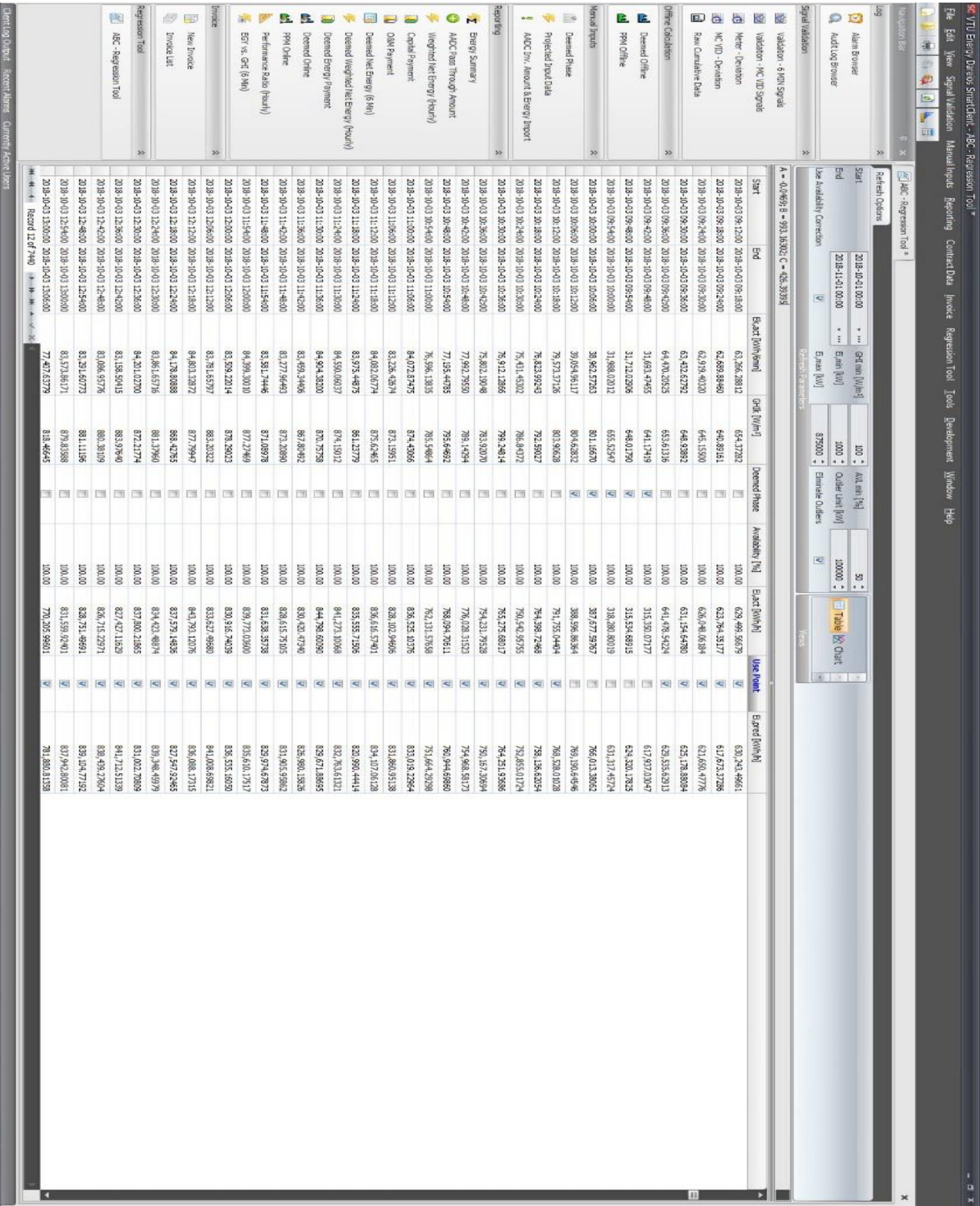


Figure 4: Report view of the Regression Tool for Deemed Energy Coefficients A, B, and C



3.1.2.6 Dareios SC SmartClient

The data I/O for the settlement system calculations are accomplished through the Dareios SmartClient. The Dareios SmartClient software can be installed on an unlimited number of computers in the plant network.

3.1.3 Software Licenses and User Interface

The following Table provides an overview of the software licenses provided in the offered system.

	PASS
Dareios SmartClient	✓
DB Database	✓
CN Calculation Node	✓
RGM Report Module	✓
IGM Invoice Module	✓
RT Regression Tool	✓

Table1 : Overview of Software Licenses

The Dareios Client software will be provided through ENEXSA’s download portal for installation at the end user’s computers at site. All other software will be designed, programmed, installed and configured on the system server computer by ENEXSA’s engineers.

The PPM/PASS system at the site as well as the system at the offtaker will each be equipped with softkey licenses for PASS.

For the PASS system installed at site the total number of concurrent users accessing the system via the Dareios SmartClient is three (3).

The copy of the PASS system at SPPC offices will include a license for one (1) concurrent user of the Dareios SmartClient. Figure 5 shows a sample invoice input form of the Dareios SmartClient. The left-hand side bar shows the frequently used functionalities and reports of the PASS system, which are:

- Signal validation,
 - Reporting,
- Off-line calculation,
 - Invoice generation.



VTU Energy Dareios SmartClient - sys://etc/Invoice/PWPA.niv

File Edit View Signal Validation Offline Calculation Reporting Contract Data Invoice Tools Window Help

Navigation Bar

Log

Alarm Browser

Audit Log Browser

Log Browser

Signal Validation

PASS Signal Validation

PASS Signal Overview

PASS Metering Raw Data

FDM Signal Validation

FDM Signal Overview

FDM Metering Raw Data

FDM Result Validation

Offline Calculation

Reporting

Declaration & Availability

Power Summary

Water Summary

Non-Conforming Water

Gas Summary

Backup Fuel Summary

Startup

FDM Online Calculations

CCM Online Calculations

Power Capacity Charge

Energy Charge

Water Capacity Charge

Water Charge

Fuel Charge

Contract Data

Billing Period

Contract Period

Static

User Inputs

Invoice

New Invoice

Invoice List

PASS - Overview

Invoice

Alarm Browser

FDM Offline Calculations

FDM - Scenario Point Verification

Billing Period Data

From:

Generator
P.O. Box 54311
Production Site

To:

Off-taker
P.O. Box 12345
Electricity & Water City

ID:

b6af882b-5c4a-4738-86af-f795a0a69449

Serial Number:

Start:

2012-01-01 00:00:00

End:

2012-02-01 00:00:00

Notes:

Payment Details:

Please pay the above amount within 45 days of receipt of invoice to:
Bank name: Bank Name
Account title: Account Title
Account No.: 123123 123 456
Swift Code: 123456

Financial Officer:

Financial Officer

Inputs

Ready

10.11.255.123

1.1.0.1446

Figure 5: Sample Invoice Input dialog (Dareios SmartClient) – example from Sakaka project.



3.2 Offshore KSA and Remote Services

3.2.1 Project Management

A Project Manager (PM) will be assigned by ENEXSA/MEMFITS to ensure all project requirements are met. The Project Manager coordinates all activities required for the project.

a. Project Scheduling – The PM will supply a project schedule for all deliveries of services and documentation.

b. Project Coordination – The PM makes sure that all aspects of a project are completed per the project schedule and coordinates all project activities with the appropriate members of Buyer's project team.

c. Single Point of Contact – The PM is the single point for contact for all aspects of the project. The Project Manager coordinates with the appropriate expertise to resolve all issues.

d. Documentation – The PM submits periodic project status reports, minutes of all project meetings which he attends and project-specific documentation.

3.2.1.1 PASS Documentation

In addition to the extensive PASS on-line help included in the system, the following documentation will be provided (in English language) as part of the PASS installation:

- a. Functional Design Specification (including hardware specifications)
- b. PPM/PASS Administrator Manual
- c. Third party documentation (as received from the vendors)
- d. FAT and SAT documentation including all test reports

All documentation will be provided in electronic format (pdf) or as received by third party vendors.

3.2.2 Functional Design Specification

The Functional Design Specification for the PASS system (PASS FDS) will be the first deliverable of ENEXSA to MEMFITS. Based on the requirements of the PPA, the Functional Design Specification will define the design features of the PPM and the Settlement System. The FDS documents will be discussed in detail and agreed with MEMFITS's project manager during the design phase of the project.

If at a later point of time during the development of the software certain specifications need to be modified due to formerly unknown facts, a revised specification or 'work around solution' will be developed and agreed with MEMFITS, and the respective changes in design and scope will amend the FDS

3.2.2.1 Remote FDS Discussion with Project Company and SPPC

The draft of the Functional Design Specification (FDS) shall be discussed in two half-day web meetings,

1. with the project company, and
2. subsequently with the project company and the offtaker Saudi **SPPC**.

3.2.3 Remote Support of PASS Factory Acceptance Test

The functionality of the individual modules of the PASS system will be demonstrated at the FAT based on a test procedure developed and agreed with MIMFITS.

For the PASS system, data validation and storage of raw and processed data as well as the integration between the plant system and the PASS copy and CSS databases as per SPPC specifications shall be demonstrated during the FAT. MEMFITS will be responsible for the respective hardware setup (including mimicking the Replica Server and CSS database servers) to demonstrate the data exchange in line with the structure as shown in Figure 1 of this Proposal or any deviations that have been mutually agreed in the FDS. ENEXSA will support the preparation of the FAT by uploading and testing the software through remote connection prior to the FAT.

ENEXSA will support the Factory Acceptance Test at the MEMFITS's workshop through web meetings and remote access to the systems supplied by ENEXSA. Utilizing a pre-populated database for the respective sample period, the process of generating the invoice as per the PPA will be demonstrated. ENEXSA's presence through remote connection is limited to a total of 16 hours.

3.2.4 Support Services Offshore KSA for the Off-line Audit

ENEXSA will provide telephone/email support for clarifications during the PASS audit until the successful completion of PASS & PPM audit by a third-party auditor. It is recommended to invite the auditor to the FAT (see Section 3.2.3 above).

ENEXSA will use the PASS at ENEXSA offices to produce data for the evaluation according to the instructions of the auditor. It is highly recommended to agree on a common format for specifying the scenarios to avoid that data are being interpreted differently by the parties. It has proven to be very efficient that ENEXSA provides templates for specifying the scenarios.

The auditor shall define up to fifteen (15) invoice scenarios to test the PPM and PASS. ENEXSA shall create the input data files for PASS and run these files through the system to create the monthly invoices. ENEXSA shall provide the test invoices to the Auditor for evaluation.

3.2.5 Remote PPM/PASS Commissioning

System installation and commissioning on the plant site shall be performed with the below details:

- a. Schedule: to be determined
- b. Location: remote connection to the PPM/PASS servers at the plant site
- c. Estimated Duration: **four (4) days**

3.2.6 Remote PPM & Settlement System Training

A combined PPM & PASS training course for SPPC staff, the plant accounting department and the system administrators will be held at the plant site.

- a. Schedule: prior to commissioning
- b. Location: web meeting. It is recommended to limit the number of attendees to eight.
- c. Estimated Duration: Two (2) training sessions with two (2) days each plus one (1) day of PASS workshop session, i.e. a **total of five (5) days**
- d. Topics: PASS training topics will include a detailed introduction into the calculation procedures, data handling modules, reporting tools. Database backup and recovery for the system administrator will be covered.

Please note that the PASS Technical Support Agreement includes preferred rates for training sessions and advanced user-training classes, so that both, general use and on-site expertise can be maintained over the entire contract term.

3.2.7 Remote PPM/PASS Commissioning at SPPC CSS

System commissioning at SPPC CSS shall be performed at the same time as the commissioning of the plant system in order to establish and test the required on-line connections.

- d. Schedule: to be determined (same as on-site commissioning)
- e. Location: remote connection to the PPM/PASS server at the CSS
- f. Estimated Duration: **four (4) days**

3.2.8 PASS Remote Site Acceptance Test (PASS RSAT)

The Site Acceptance Test (SAT) shall be performed at the IPP subsequent to commissioning.

- a. Schedule: to be determined
- b. Location: remote connection to the PPM/PASS server rack at the plant site
- c. Estimated Duration: one (1) day
- d. Topics:
 - Demonstration of PPM
 - Demonstration of PASS Report Generation Module
 - Demonstration of Manual Data Input Module
 - Invoice Generation and Reporting Module

3.2.9 PASS Remote Final Testing at SPPC Offices

ENEXSA will install and commission the copy of the Plant Account and Settlement System (including Plant Performance Model) of the IPP on computer hardware supplied for the SPPC offices. For the final testing and demonstration, ENEXSA foresees the following schedule:

- a. Schedule: subsequent to commissioning
- b. Location: remote connection to the PPM/PASS server at SPPC offices
- c. Estimated Duration: **one (1) day**

System tests will be performed by comparison of the results of PASS at the plant site and PASS at CSS for identical sets of input data.

4.0 OPTIONAL ITEMS

4.1 Service Agreement and Maintenance Plan

Since we want the Buyer to receive the best value, we will include a Service Agreement, as an optional item upon request, This Service Agreement is effective for one year after the completion of the SAT. The Service Agreement is renewable on an annual basis for following years.

5.0 BUYER RESPONSIBILITIES

5.1 Plant Information Requirements

At project kick-off the following information shall be made available:

5.1.1 Plant & Equipment Descriptions

- a. All manufacturer performance curves and performance data relevant for the PPM
- b. Foreseen operating and ambient condition limits
- c. Any operational limits

5.1.2 PPM Measured Data Requirements

The PPM requires the following process tags (direct measurement or all individual measurements to calculate the respective variables) to be available and accessible throughout the various phases of the project (installation, model testing, supporting service contract duration):

- a. global horizontal irradiation,
- b. global solar irradiation on plane of array,
- c. module temperature, and
- d. Net Electrical Energy
- e. Plant Availability (as a percentage of nominal installed capacity)

5.1.3 PPA Performance Data

- a. Guaranteed plant rating (net power output at reference conditions).
- b. All Performance data points subject to the PPA.
- c. Correlations and correction procedures as laid out in the PPA.
- d. Guaranteed data such as guaranteed performance ratios

5.1.4 PASS Measured Data Requirements

At project kick-off the following information shall be made available:

5.1.4.1 Metering System

- a. Description and documentation of the power metering system ((range limits and roll-over values, UOM);
- b. All input values relevant for the Settlement System:

5.1.4.2 PPA Settlement Data

- a. Final negotiated PPA including all tariff definitions;
- b. Billing procedures (Appendix 9 to the PPA and all documentation referred to therein);

5.2 Buyer's Support of Integration Work

5.2.1 Other Hardware and Infrastructure at the Plant Site

In case on-line support at site is required, VPN access to the PASS server computers has to be provided by Buyer.

Note: MEMFITS shall arrange his engineer based in KSA to site with necessary hardware for the necessary coordination with ENEXSA to perform the integration, configuration, testing, trouble shooting works.

5.3 Site Personnel Support and Site-Specific Logistics

For MEMFITS/ ENEXSA's on-site hardware integration works, the on-site training, the SAT and for the Technical Support Services the below items apply:

- a. MEMFITS to appoint a Project Manager responsible for obligations associated with this project. Buyer should be prepared to have accessible all required information and services to implement the system as described in this document.
- b. MEMFITS/ENEXSA requests that the person from the plant requesting assistance be fully trained on the MEMFITS/ENEXSA supplied software programs. Additionally, the hardware/software systems shall be available, with appropriate permissions and rights, and accessible (Ethernet or modem) during any tech support call to MEMFITS/ ENEXSA.
- c. Customer will provide all plant services such as light, heat, water, electric power and disposal of all waste materials.

5.4 Safety

a. MEMFITS/ENEXSA will furnish personal safety equipment that meets the needs of performing the scope of work described herein, such as hard hats, high visibility vests and safety boots. Other mandatory personal safety requirement specific to the plant is not part of this scope of supply. It is assumed this will be provided by Buyer.

b. Buyer shall perform testing to make the work area free from hazardous materials. It is assumed that no hazardous material will be encountered in performing the scope of work. Disposal or abatement of any hazardous or regulated material will be the responsibility of Buyer.

c. All safety training if required is in the Buyer scope of supply.

6.0 Clarifications and Assumptions

6.1 Settlement Procedure

Changes to the Settlement procedures during the project execution phase which require rework/reconfiguration of the PASS system will also require a change order.

6.2 Data Simulation

If on-line data are not available at the time of Site Acceptance Testing, then the Site Acceptance Test shall be performed using simulated data of the FAT.

7 Exceptions and Clarifications to the Specifications

Reference is made to the document “Central Settlement System – Standard Interface Specification for Data Exchange Solar Plants, Version 6, March 2023” (CSS Specification v6).

7.1 Data Storage in Binary Format

CSS Specification v6 Sections 3.1 / 8.	ENEXSA Comment/Clarification
All outputs of the offline PSS shall be calculated in the offline PSS server solely, and should be stored in SQL servers only, in readable blain text format. These data shall not be encrypted or stored in binary format.	While all settlement data (inputs and results) will be stored in plain text format, ENEXSA takes exception for the storage of the system configuration and produced invoice files. Similar to MS Office files (Word, Excel, Powerpoint etc.) these files do not pose any harm to the CSS systems, as they are not self-executing.

7.2 Trial/Scenario Feature

CSS Specification v6 Section 6	ENEXSA Comment/Clarification
Trial/scenario Feature: The Offline version has extra feature which is the ability to allow the user to run mockup scenario with custom raw inputs. This feature shall not impact the offline version hourly process as in the online version.	Since the Offline PSS must be strictly kept synchronized with the Online PSS at the plant site, it is technically impossible to include this trial/scenario feature in the Offline PSS.



7.3 Operation without Any Human Interaction

CSS Specification v6 Section 6	ENEXSA Comment/Clarification
Both versions shall run without any human interaction in systemically basis to generate all the calculations according to the PPA.	The PASS processes input data on-line without any human interaction, but it does require specific user input data for generating the monthly invoice. The nature of the contract itself requires certain calculations (invoice, determination of A,B,C coefficients) to be triggered by a user after providing the required manual inputs.

7.4 Version Control of Manual Inputs

CSS Specification v6 Section 7.2	ENEXSA Comment/Clarification
However, if there is a need for adjusting the data automatically or manually, the PSS shall have a different column from the original data column. It should be called "adjustment" column, where modified value is recoded. Moreover, any new modification in adjustment column, which overwrite an old value within the column, must be recorded while the old value is kept/recorded in form of an old version.	It is not relevant for the invoice which values may have been used as adjustment prior to the generation of the final submitted invoice. Adding version control to manual inputs for signal validation will thus not be of practical value but would add significant complexity and cost to the settlement system. Therefore, ENEXSA takes exception to this feature.

7.5 Linux (Cross Platform)

CSS Specification v6 Section 8	ENEXSA Comment/Clarification
The PSS shall be executable (natively) in Windows and Linux (cross platform).version.	The ENEXSA PASS executes exclusively on industry standard Windows Server operating systems. ENEXSA takes exception for execution on Linux O/S.

**Acquisition Management System (AMS)
For ArRass2 Solar PV IPP in Saudi Arabia**



■ 1.0 EXECUTIVE SUMMARY

MEMFITS would like to thank **Saudi Power Procurement Company (SPPC)**. for providing us an opportunity to submit a proposal towards Acquisition Management System (AMS) And PASS Hardware for wind power plants with a total net capacity of 1800 MWe which will be located in different regions of Saudi Arabia..

This document describes the general and specific services required to develop , design and supply the Acquisition Management System (AMS) And Plant Accounting & Settlement system (PASS) Hardware with necessary integration works and project management services .

■ 2.0 PROPOSAL STRUCTURE

This proposal is based on the following documents:

- I. Technical specification for Settlement System
- II. Annexures

□ This offer includes

- AMS system software, with the necessary licenses to covers the planned meters and the needed multiple data points,
- The AMS hardware to run the system,
- PASS Hardware to run the system, (At plant side & SPPC office).
- Integration Services.
- Project Management Services.
- Engineering , Design , Manufacturing , Testing and commissioning.
- Training and documentation.

3.0 WIND Farm - DETAILS

Project Name	Solar power plants - Ar Rass 2
Location	Ar Rass 2 located in the Riyadh province, Saudi Arabia
Type	Solar Power Plant Field
Capacity	2000 MW

Project Requirements

- Redundant Meter Data Acquisition System (AMS System with Hardware , Software & Services).
- The AMS will be used for the daily collection of all Plant settlement data from all meters (Energy meters & Fuel meters). All electronic data relating to the settlement process will be stored in secure manner.
- Plant Accounting Settlement System (PASS) with Hardware , Software and Services.
- Tariff calculation Module in accordance with Gas Field
- MDA shall be connected to DCS via redundant OPC link
- Browsing, printing and managing of invoices
- Billing and invoicing calculations
- Monthly invoice generations (Capacity charge, Output charge and other charges)



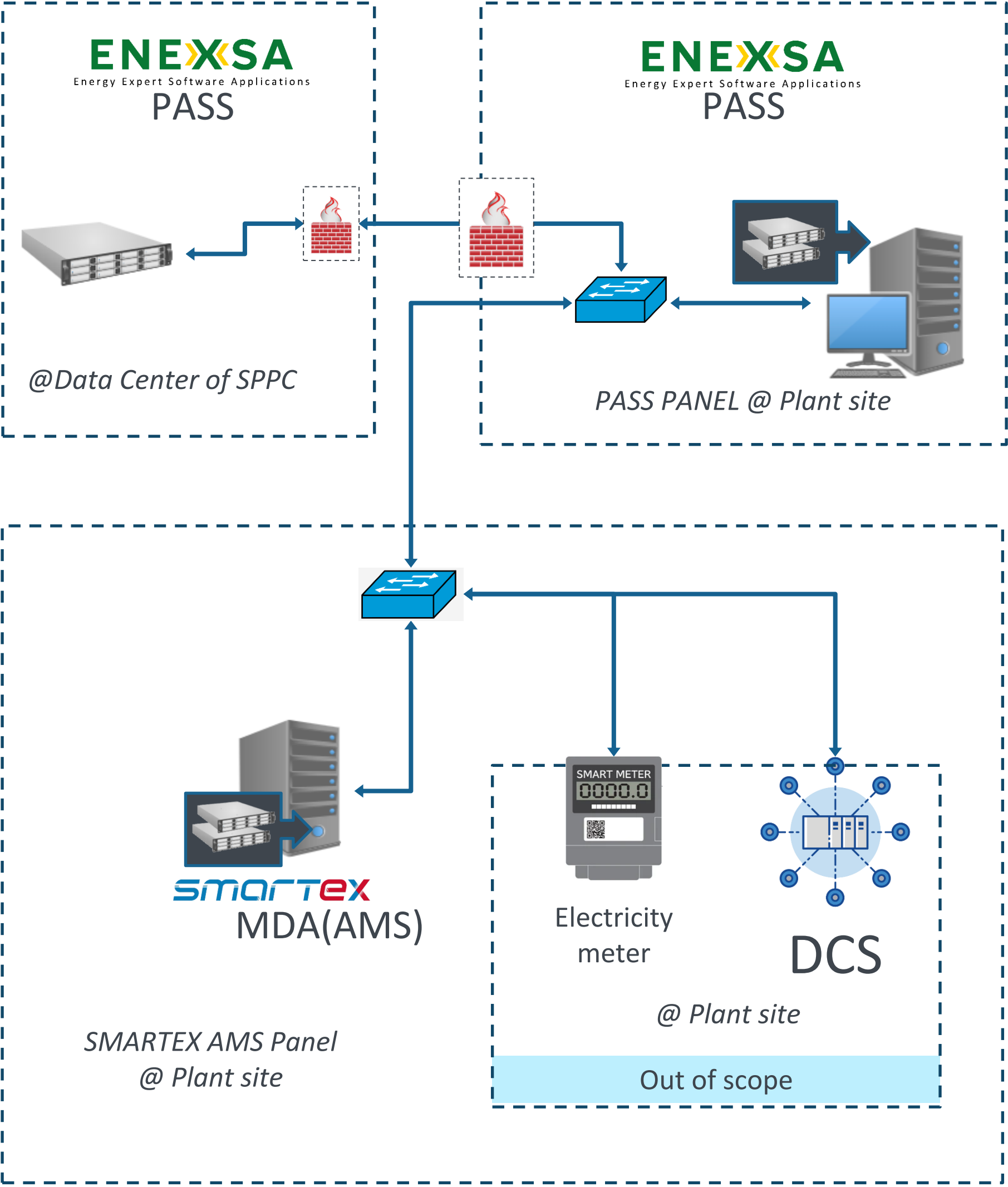
4.0 PROPOSED SOLUTION

MEMFITS offers to provide End – To – End solution includes the Acquisition Management System (AMS) , Plant Accounting & Settlement system (PASS) Hardware and project management services for MEMFITS Scope , to meet the required functionality of the plant settlement .

5.0 SYSTEM CONFIGURATION

Our SMARTEX AMS system is a platform for Sewage Water Plants capable providing complete insights of generation plant solutions including the settlement system.

Network Topology



6.0 SCOPE OF WORK

6.1 Supply of One (1) Acquisition Management System (AMS) Panel includes the System Server with necessary hardware and software as illustrated in above picture .(One at Plant Site) with Redundant Server

6.2 Supply of Two (2) PASS system Hardware .(One at Plant Site & One at SPPC office)

6.3 Design , Engineering, Manufacturing and Assembling , Testing & Inspection, Packing & Forwarding.

6.5 Installation Supervision, System Configuration and the integration .

6.6 Site Testing, Commissioning.

6.7 Training to the site team and documentation

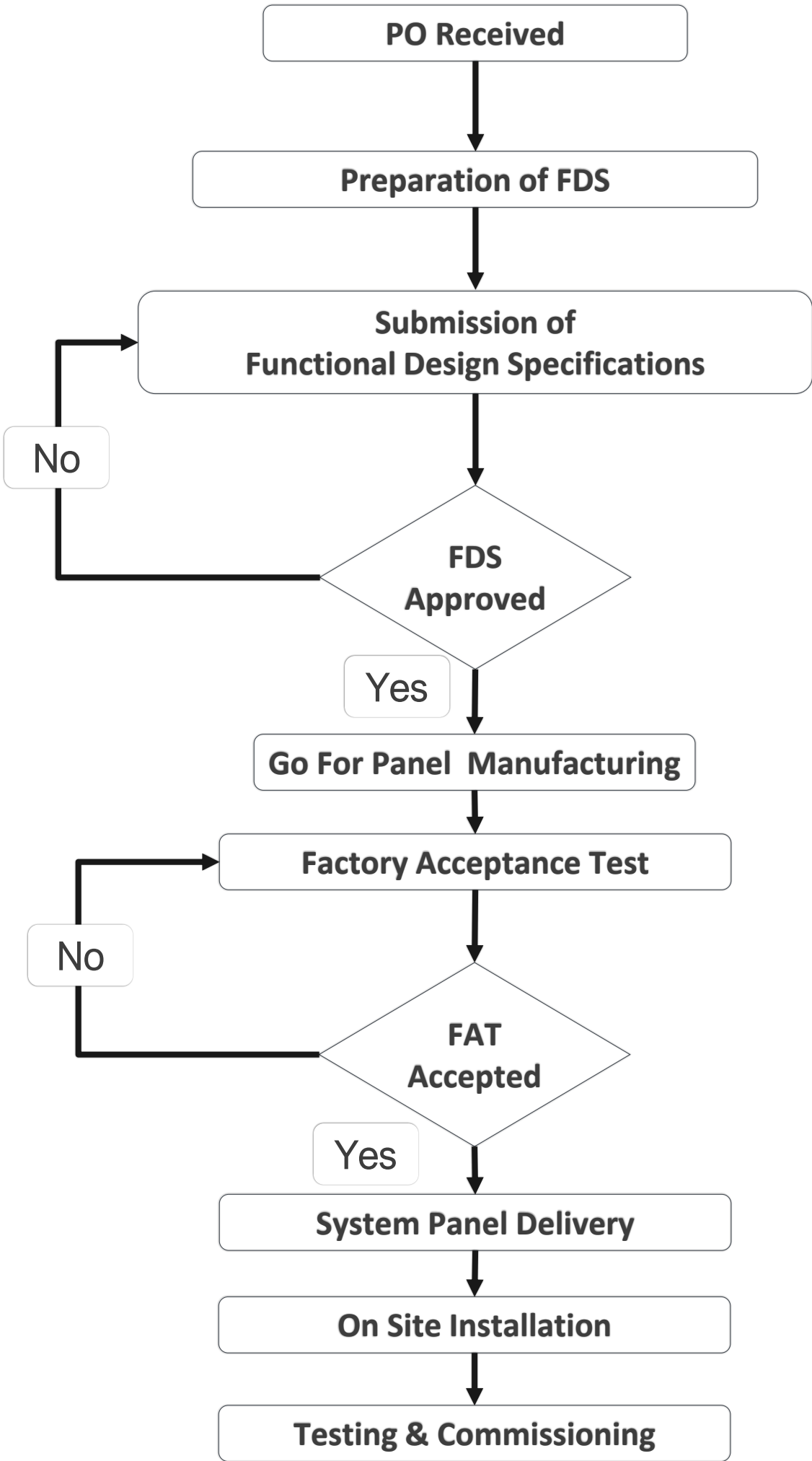
6.8 Additional Services : In addition to the services to define, develop , configure and integrate the above described scope of work the following services are included in the scope of supply.

6.0 SCOPE OF WORK

- a. Project Management –** The Project Manager will be assigned by MEMFITS to ensure all project requirements are met. The Project Manager coordinates all activities required for the project. Any activity that is necessary for the successful implementation of the project is under the responsibility of the Project Manager.
- b. Integration Scheduling –** MEMFITS Project Manager will supply a project schedule for all deliveries of services and documentation.
- c. Coordination –** MEMFITS Project Manager will make sure that all aspects of the integration are completed per the Integration schedule and coordinates all the integration activities with the appropriate members of the client project team.
- c. Site Acceptance Test (SAT) .**
- d. Commissioning .**
- e. Documentation –** MEMFITS Project Manager will submit periodic status reports, minutes of all meetings which he attends and project-specific documentation.

6.0 SCOPE OF WORK

6.1 EXECUTION METHODOLOGY



6.2 BUYER RESPONSIBILITIES

1. All on-site integration works shall be performed by MEMFITS engineers , Prior to the set- up of the New Hardware required for the integration work, the Buyer shall supply a network connection , required IP address and power supply at the specific location .
2. The Buyer Shall provide all plant services such as light , electric power and disposal of all waste materials.
3. The Buyer must provide the required approval from the local utilities and any other work permits prior to send our team to the plant site.

Thank you,
We are looking forward to working with you.

